

FOUNDATIONAL QUANTS _22-9-26

Directions for question 1: Type in your answer in the input box provided in the question.

$$\frac{(19.9)^3 - (18.1)^3}{(19.9)^2 + (19.9 \times 18.1) + (18.1)^2} =$$

Solution:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$\therefore \frac{a^3 - b^3}{a^2 + ab + b^2} = a - b$$

$$\frac{19.9^3 - 18.1^3}{19.9^2 + 19.9(18.1) + 18.1^2} = 19.9 - 18.1 = 1.8.$$

Ans : (1.8)

Directions for question 2: Select the correct alternative from the given choices.

A number consists of two digits. The digit in the unit's place is twice that in the ten's place. If the digits are reversed, and the new number is increased by 6 it equals two times the original number. Find the number.

- ☐ a) 36
- ☐ b) 24
- ☐ c) 42
- ☐ d) 28

Solution:

Let the tens digit and units digit be x , $2x$ respectively. $\therefore 2(10x + 2x) = 10(2x) + x + 6 \Rightarrow 3x = 6 \Rightarrow x = 2$. The number is 24.

Choice (B)

Directions for question 3: Type in your answer in the input box provided below the question.

Among three partners A, B and C, A and B together invested a sum equal to the capital of C. If they made a total profit of Rs.36,000 after one year, find C's profit.(in Rs.)

Solution:

When the volume is 50 units, the density is 48 units.
 $\therefore$ When the volume is 75 units, the density is 32 units.

Choice (B)

The cost of some furniture inclusive of sales tax is Rs.7500. If the sales tax is 25%, find the original cost of the furniture.

☒ a) Rs.6000 (Your answer - ✓)

☐ b) Rs.5625

☐ c) Rs.5000

☐ d) Rs.5500

Solution:

Let the cost (without sales tax) be $4x$. Cost (including tax) = $4x + x = 7500$
 $\Rightarrow x = 15000$.
 $\therefore$ Cost (without tax) = $4x = 6000$

Choice (A)

For what values of x is $25^x + 1$ divisible by 13?

☐ a) All values of x .

☒ b) Odd values of x . (Your answer - ✓)

☐ c) Even values of x .

☐ d) Cannot be determined

Solution:

The first few values of x and the corresponding value of $25^x + 1$ are tabulated below.

x	$25^x + 1$
1	26
2	626 this is not divisible by 13
3	$25^3 + 1 = (25 + 1)(25^2 - 25 + 1)$ This is divisible by 13.

$a^n + b^n$ is divisible by $a + b$, when n is odd. $\therefore 25^x + 1$ is divisible by 26 (and hence by 13) for odd values of x .

Choice (B)

Directions for question 7: Type in your answer in the input box provided below the question.

One-third of one half of a number is subtracted from half of the number. This difference is again reduced by one-fourth of the number. If the result is 25, then what is the number?

Your Answer: Unattempted

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Solution:

One third of half is $\frac{1}{6}$. This is subtracted from 'the other' half. (In this context, it does not matter from which half it is subtracted). The result is $\frac{x}{2} - \frac{x}{6}$, where x is the given number. This is decreased by $\frac{x}{4}$.

$$\therefore \frac{2x}{6} - \frac{x}{4} = 25 \Rightarrow \frac{4x}{12} - \frac{3x}{12} = 25 \Rightarrow x = 300.$$

Ans : (300)

When an article is sold for Rs.1,500, the profit earned is one and half times of the loss incurred when it was sold for Rs.500. Find the cost price.

- ☐ a) Rs.1,200
- ☐ b) Rs.1,000
- ☐ c) Rs.900
- ☐ d) Rs.1,150

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Solution:

If the article is sold at 500, let the loss be 2x. If the article is sold at 1500, the profit is 3x.

$\therefore 500 = c - 2x$, $1500 = c + 3x$, where c is the cost price.

$\therefore 1000 = 5x \Rightarrow x = 200$ and $c = 900$.

Choice (C)

Two companies have their employees in the ratio 19 : 16 and their average wages in the ratio 4 : 5. What is the ratio of the wage bill of the two companies?

- ☐ a) 39 : 60
- ☐ b) 12 : 13
- ☐ c) 17 : 18

☒ d) 19 : 20 (Your answer - ✓)

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Solution:

In one company the number of employees is $19x$ and their average salary is $4y$.
In the other company, the number of employees is $16x$ and the average salary is $5y$.
 $\therefore$ The ratio of the total wages is
 $19x (4y) : 16x (5y) = 19 : 20$

Choice (D)

Find the units digit of $7^{48} \times 3^{56} \times 165^{35}$.

☐ a) 0

☒ b) 5 (Your answer - ✓)

☐ c) 1

☐ d) 7

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Solution:

Consider each of the numbers and find their units place and Units place of 7^{48}
Cycle of 7 is 7,9,3,1 i.e., period is 4
 $48/4 = 12$, no remainder i.e. 12 cycles
 $\therefore$ units place of 7^{48} is 1.
Units place of 3^{56}
Cycle of 3 is 3,9,7,1 i.e., period of 4
 $56/4 = 14$, no remainder i.e. 14 cycles.
 $\therefore$ units place of 3^{56} is 1
Units place of 165^{35}
Units place of 5 raised to any power is 5.
 $\therefore$ Units place of 165^{35} is 5
 $\therefore$ Units place of $7^{48} \times 3^{56} \times 165^{35}$ is $1 \times 1 \times 5 = 5$

Choice (B)

One-fifth of the cost price, one-seventh of the marked price and one-sixth of the selling price are all equal. What is the gain or loss to the trader?

- ☐ a) 20% gain
- ☐ b) $16\frac{2}{3}\%$ loss
- ☐ c) $14\frac{2}{7}\%$ gain
- ☐ d) 10% loss

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Solution:

Let the cost price, selling price, marked price be c, s, m respectively. $\frac{c}{5} = \frac{s}{6} = \frac{m}{7} = x(\text{say})$
 $\therefore c = 5x, s = 6x, m = 7x$. Profit percentage = $\frac{6x - 5x}{5x} (100\%) = 20\%$
Choice (A)

In a two-digit number, the sum of the digits is nine. If the digits are reversed, the number obtained is $\frac{8}{3}$ times the original number. The number is ____.

☐ a) 45

☒ b) 27 (Your answer - ✓)

☐ c) 63

☐ d) 54

Solution:

Let the number be ' ab ' = $10a + b$.
The reverse is $10b + a$.
 $10b + a = \frac{8}{3} (10a + b) \Rightarrow 30b + 3a = 8a + 8b$
 $\Rightarrow 22b = 5a \Rightarrow 2b = \frac{5}{2}a$. The only possibility is $b = 5, a = 2$. We don't even need the data $a + b = 9$
The number is 25.
Choice (B)

A trader bought 15 books at the price of 12 books and sold 12 books at the price of 15 books. What is his percentage of profit per book sold?

- ☐ a) 50%
- ☐ b) 25%
- ☐ c) 20%
- ☐ d) 56.25%

Solution:

In the phrases 'price of 12 books' and 'price of 15 books' the word price refers to the same number. We can think of this as some reference price. There are suppliers who are willing to supply books to the trader at a lower prices and there are customers who are willing to buy from the trader at a higher price. Let this reference price per book be a . The number of books (n), their cost price for the trader (c) and their selling price for the trader (s) are tabulated below.

n	c	s
15	$12a$	
12	$9.6a$	$15a$

The profit percentage is $\frac{5.4a}{9.6a} (100\%)$

$$= \frac{9(100)}{16} \%$$

$$= 56.25\%$$

Choice (D)

Singh and Raju invest Rs.20,000 and Rs.25,000 respectively to start a business. Raju withdraws after 10 months and Singh gets Rs.12,000 as his share after one year. Find Raju's share.

- ☐ a) Rs.24,500
- ☐ b) Rs.14,500
- ☐ c) Rs.12,000
- ☐ d) Rs.12,500

Solution:

The data is tabulated below.

	Singh	Raju
Capital	20	25
Period	12	10
Profit share	48	50

A Singh's share is actually 12,000, Raju's is 12,500

Choice (D)

Find the number of factors of 1728.

☒ a) 28 (Your answer - ✓)

☐ b) 18

☐ c) 36

☐ d) 26

Solution:

$$1728 = (12)^3 = 2^6 \times 3^3$$

$$\text{Number of factors} = (6 + 1)(3 + 1) = 28$$

Choice (A)

A trader sold an article and got one-fourth of selling price as profit. Find the percentage of profit.

☒ a) 25% (Your answer - ✗)

☐ b) $33\frac{1}{3}\%$ (Correct Answer - ✓)

☐ c) 20%

☐ d) $26\frac{2}{3}\%$

In a two digit number the digit in the ten's place is three more than the digit in the unit's place. Two times the digit in the unit's place is equal to the digit in the ten's place. Find the number.

☐ a) 52

☐ b) 74

☒ c) 63 (Your answer - ✓)

☐ d) 41

Solution:

The tens digit is double the units digit and it is 3 more than the units digit. The number is 63. Choice (C)

A man spends one-fifth of his income on rent, one-fourth on food, half on other expenses and saves the remaining. What fraction of his income does the man save?

☐ a) $\frac{1}{7}$

☐ b) $\frac{2}{15}$

☐ c) $\frac{1}{10}$

☒ d) $\frac{1}{20}$ (Your answer - ✓)

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Solution:

The man spends $\frac{1}{5}$ on rent, $\frac{1}{4}$ on food, $\frac{1}{2}$ on other expenses. i.e., he spends $\frac{4}{20}, \frac{5}{20}, \frac{10}{20}$ on the 3 items. He saves the remaining $\frac{1}{20}$ Choice (D)

Directions for question 19: Type in your answer in the input box provided below the question.

Out of four persons P, Q, R and S, P and S together earn Rs.7,000. If the ratio of earnings of P, Q, R and S is 2 : 3 : 4 : 5, find R's earnings. (in Rs.)

Your Answer: 4000 ✓

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Solution:

The ratio of the earning of P, Q, R, S is 2 : 3 : 4 : 5. P and S together earn Rs 7000.
As 7 parts is 7000, 4 parts is 4000. Ans : (4000)

Number of ways in which 84 can be written as a product of two coprimes is _____.

☒ a) 8 (Your answer - ✕)

☐ b) 12

☐ c) 4

☐ d) 2

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Solution:

$$84 = 4 (3) (7)$$

$$\therefore 84 = 1(84)$$

$$= 4(21)$$

$$= 3(28)$$

$$= 7(12)$$

84 can be written as the product of 2 co primes in 4 ways.

Choice (C)